USER GUIDE

Valencia Stock Energy Workbench

How to load a city's data, start a simulation, and read every number that comes out — screen by screen, in the order you will use them.

01 · FILES
Four inputs
Validate and activate the data the engine will read.

→

02 · RUN
One model per building
A full simulated year, for every building in scope.

→

03 · OUTPUTS
Results and evidence
Totals, per-building ledger and the raw output files.

Normative in origin
Lighting and appliances follow a standard density per m², hot water a standard draw per person — 90 % of the total.

HVAC is simulated
Heating and cooling come from real building physics over 8,760 hours.

Verified, not validated
Scored against a published reference model. Never compared with a meter.

Three screens, in this order
Files → Run → Outputs

Example run throughout
Benicalap district · `benicalap_v8`

Runs on your own machine
`http://127.0.0.1:8765`

Engine
EnergyPlus 25.2.0 · OpenStudio 3.11.0

1 Start here

Everything runs locally. The browser is only the interface — no data leaves the machine, and closing the tab does not stop a run.

1. **Install once.**
   From the unpacked project folder: `./scripts/install.sh` on macOS or Linux, `.\scripts\install.ps1` in PowerShell on Windows. It builds the interface, checks that EnergyPlus 25.2.0 really executes, and stops if the toolchain is wrong instead of failing later mid-run.
2. **Start the application.**
   `./scripts/start.sh` or `.\scripts\start.ps1`. It opens at `http://127.0.0.1:8765/#/files`. Keep the terminal window open — that process is the engine.
3. **Look at the top-right of the screen before anything else.**
   Files shows **RUN READY** or **INPUTS REQUIRED** with the missing item named. Run and Outputs show the verified profile fingerprint — the identity of the physics that will produce your numbers.
4. **Stop with a single `Ctrl+C`.**
   Every finished building is already written to disk, so an interrupted run is resumed, not restarted.

> **You can explore before you simulate anything**
> The product ships with a completed district run, so Outputs is fully populated on first launch. Open it, click a building, open its EnergyPlus file — all of §4 works before you have run a single simulation of your own.

> **If a screen refuses to do something**
> It is a gate, not a fault. The engine records what it cannot support instead of inventing a default — a rejected file names the column it lacked, and a blocked run names its missing input. Read the message; it is the diagnosis.

2 Files — load the four inputs

A run needs four things: what the buildings are, how much of them is housing,
what the weather does, and which library of constructions to build from.
Nothing starts until all four are active.

1. **Press Upload on a card and pick the file.**
   It is checked first and only becomes active if it passes. A file that fails is refused with the reason — it never becomes a silent partial input.
2. **Or choose an already-validated file from the card's dropdown.**
   Every file you have ever uploaded stays selectable, so you can switch between two GIS versions without re-uploading either.
3. **Check that all four cards read ACTIVE.**
   A card still marked **REQUIRED** is what is holding the run back. The climate card needs both halves — the annual EPW and the design days — before it turns.
4. **Confirm the header says RUN READY, then go to Run.**
   That badge is the engine's own answer, not a UI guess: it resolves every input and every entry point it will need.

GEOMETRY + IDENTITY

Building GIS

The cadastre itself: one row per parcel, carrying the shape, the height, the typology class and the people.

FIELDS THE ENGINE READS

`refparcela · altura_max · cluster`
`pob_total · num_vivend · nombre · geometry`

GATE Required fields present, geometry readable, coordinate system recorded.

RESIDENTIAL AREA + USE

Tipo15 companion

The cadastre's floor-by-floor record: how much dwelling area exists, on which floor, for what use.

FIELDS THE ENGINE READS

`31_pc (joins to refparcela)`
`252_planta · 428_uso · 442_sup_Residencial`

GATE All four columns parseable; the join to parcel references is measured and reported, not assumed.

ANNUAL + SIZING WEATHER

Climate pair

Two files that must agree with each other: the year that is simulated, and the extremes the equipment is sized on.

WHAT EACH ONE DRIVES

`.epw` — every one of the 8,760 simulated hours
`.ddy` — heat-pump capacity in stage 8

GATE A complete 8,760-hour year; a winter and a summer design day. The two are cross-checked as one pair.

OPENSTUDIO SOURCE

PlantillaOS template

The library every model is assembled from. It carries no geometry — the geometry is built per building.

WHAT IT SUPPLIES

constructions and materials · schedules · space types
equipment definitions · thermostat set points

GATE Every construction, schedule and space type the engine will ask for must already exist — and no geometry.

Nothing is hard-coded into the program. A newer cadastre, a different weather year or a modified template can be swapped in from this screen without touching any code — provided it passes its gate.

What each file must contain to be accepted

INPUT	ACCEPTED	THE GATE IT HAS TO PASS
Building GIS	`.gpkg` `.shp` `.geojson` `.zip`	Must carry `refparcela`, `altura_max` and geometry. The typology field and the district names are read from it — those are what the Run screen offers you as scopes. A shapefile must be uploaded with its sidecars.
Tipo15 companion	`.csv`	Semicolon-delimited, Latin-1, containing `31_pc`, `252_planta`, `428_uso` and `442_sup_Residencial`. It is refused if no parcel reference or no positive residential area survives parsing.
Annual EPW	`.epw`	At least 8,760 hourly rows — a partial year is refused. The site header is read, and whether the file carries design conditions is recorded.
Design days (DDY)	`.ddy`	At least one `WinterDesignDay` and one `SummerDesignDay`. The standard 99.6 % heating and 0.4 % cooling days are then selected by name, not guessed.
PlantillaOS template	`.osm`	All twenty-four named roles the chain binds to — space types, constructions, the eight materials, thermostats, hot water, window frame, blind — must be present. It is a library, not a building: the geometry is built from your GIS, never taken from here.

An active card shows its evidence underneath: the **snapshot checksum**, the contract it satisfied, the row count, the coordinate system, and the managed path the file was copied to. That is what you quote when someone asks which data a result came from.

> **Why every file is hashed**
> Each file is copied into managed storage and its checksum recorded. A run resolves the active set once and stamps those checksums into every result row — so any row can later be traced to the exact bytes that produced it.

3 Run — start a simulation

Scope first, then the engine's own preflight, then start. The preflight is not a preview of the run — it is the same screening code the run itself uses.

1. **Choose a scope.**
   Selected buildings for one or a handful of cadastral references — the fastest way to test anything. **One district** for every eligible building in a named district. **All Valencia** for the full stock: hours of work and a large amount of disk.
2. **Fill in what that scope needs, and name the run.**
   References are typed or pasted, separated by spaces, commas or new lines. The name is what you will resume with later, so make it one you will recognise; the field shows you the filesystem-safe version it will use.
3. **Press Run preflight and read the five numbers.**
   In scope is what your choice selected. **Runnable** is what survived screening. **Excluded** is the difference — and it is itemised by reason. **Estimated time** and **estimated storage** are what the run will actually cost you.
4. **For the full city only, tick the acknowledgement.**
   It confirms you have read the exclusion count, the duration and the storage estimate. Nothing else in the product asks you to confirm anything.
5. **Press Start run.**
   Six buildings are simulated in parallel; each finished building is flushed to disk before the next begins.

IN SCOPE	RUNNABLE	EXCLUDED
1,012	**968**	**44**
every building the district selection contains	passed geometry and data screening	itemised by reason — footprint out of range, simplification tolerance, duplicate reference

Preflight for the Benicalap example, before the run that produced every other number in this guide.

While it runs

The live panel shows the percentage complete, the count of terminal records, and a running split of OK · FAILED · EXCLUDED with CPU time. Those counts are read back from the flushed ledger on disk, not from the running process's memory — so what you are watching is what has actually been written. The last 160 KB of engine output is shown underneath.

Stopping is safe, and resuming is exact

Stop safely lets the buildings in flight finish and leaves the ledger intact. **Resume same run** then continues from the first unfinished building — but only if the inputs and the physics are still identical. If either changed, resume is refused rather than producing a run that spans two versions of itself.

A failed building is not a failed run

Each building is isolated: one that fails is written to the ledger with its error and the run continues. The Benicalap example finished with 967 OK, 44 excluded and 1 failed — the failure is visible in Outputs, not hidden in a log.

4 Outputs — read, audit, export

Pick a run at the top of the screen. Finished runs and the running one are all listed; the shipped example is there from the first launch.

The six figures at the top

TOTAL SITE ENERGY **109.76** GWh / year · **modelled subset only**, never extrapolated to the buildings that were excluded	GEOMETRIC EUI **46.9** kWh/m² · divided by footprint × residential storeys	CADASTRAL EUI **50.9** kWh/m² · divided by recorded dwelling area — the comparable one
CARBON **34,723** tCO_2 / year · rests on an assumed fuel mix (see §7)	COVERAGE **95.6 %** 967 simulated · 44 excluded · 1 failed — 98.1 % by footprint area	QA STATUS **PASS** 0 quality-check failures · 0 unexplained engine errors

The small print under each figure is the basis, and it is there because it changes what the number means — §6 shows how much. Underneath, the cluster table repeats all of this per building type, with the published reference value and the difference from it in the last two columns.

Auditing one building

1. **Search the ledger.**
   By cadastral reference, by typology, or by error text. The status filter narrows it to **ok**, **failed** or **excluded**; a hundred rows are shown at a time.
2. **Click any row.**
   A panel opens with that building's status, typology, intensity, QA verdict and — if it failed — the error that stopped it.
3. **Open the files the run preserved for it.**
   Five per building, listed below. They open read-only in a new tab.

PRESERVED FILE	WHAT YOU USE IT FOR
`eplustbl.htm`	EnergyPlus's own result tables for that building — areas, loads, end uses, unmet hours. The primary evidence.
`deep_layers.json`	What each stage did to this specific building, plus the quality checks that compared the model against what EnergyPlus reported.
`model_python.osm`	The OpenStudio model as built. Open it in OpenStudio to inspect the geometry directly.
`eplusout.err`	Every warning and error the engine raised. Read this first when a building looks wrong.
`verified_profile.json`	The exact physics profile that produced the row — every locked parameter, stamped at run time.

Getting the results out

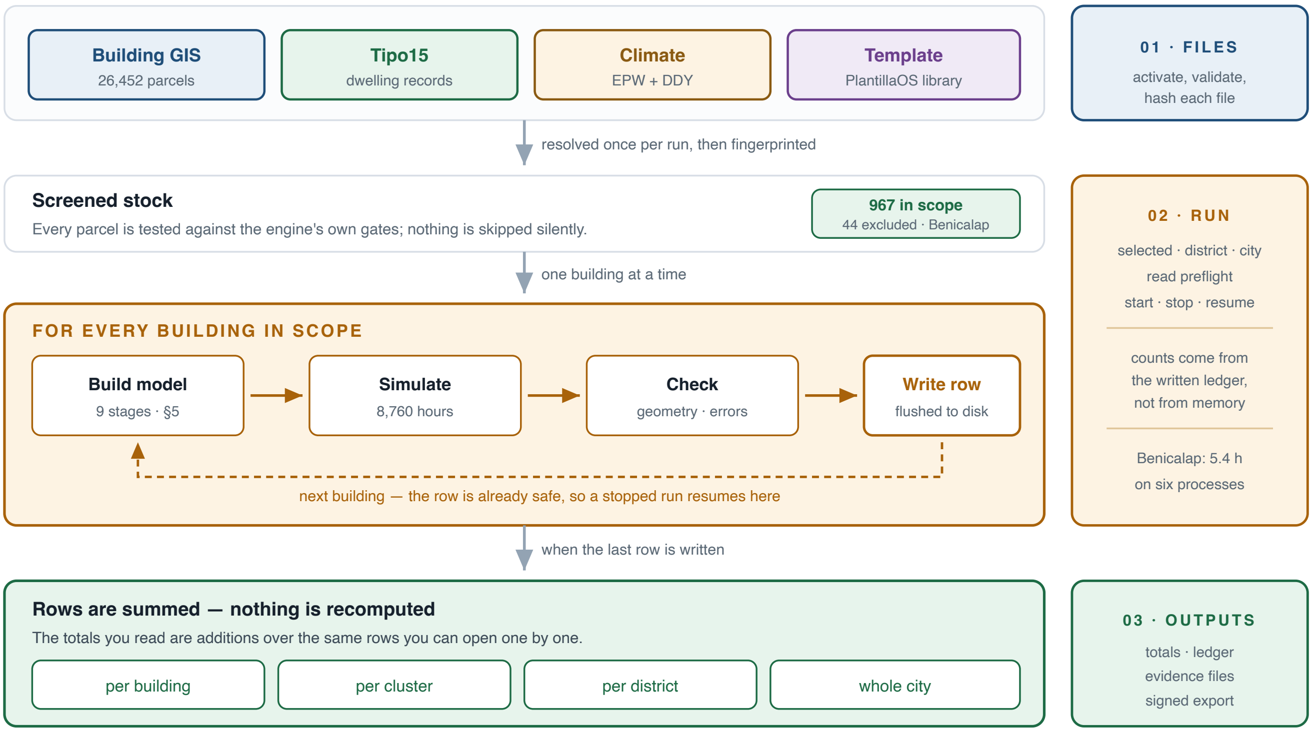
- **Building CSV** — the whole searchable ledger as one table, one row per building. This is the file to open in a spreadsheet.
- **GIS layer (.gpkg)** — the same rows with the geometry already attached, so the result opens straight in QGIS with no join to make. Drag it in and it arrives already coloured on `total_site_kwh_m2`; switch the column under **Properties → Symbology** for any of the others — heating, cooling, hot water, electricity, gas, CO_2, energy per resident, energy per dwelling. There is no separate heating file and no separate cooling file: they are columns of this one, which is why they can never disagree with each other.

The file holds three layers. `results` is one feature per building; `by_cluster` and `by_district` are the same result dissolved into typology and district polygons, each carrying its own totals, its floor-area-weighted intensity, and both a `buildings` and a `buildings_measured` count so a zone coloured on part of its stock cannot be read as if it covered all of it.
Two things are worth knowing before you total anything from it. Buildings that failed or were screened out are *in* the layer, with their `status` and `reason` and empty energies — a gap in coverage should be visible on the map, not silently missing from it. And a cadastral reference drawn as several footprints becomes *one* feature holding all of them, so summing a column gives the run's own total rather than that total multiplied by however many polygons a parcel happens to have. The file is also inside the full ZIP.

- **Heat map (.png)** — the same three quantities drawn as a picture, for a report or an email where nobody is going to install a GIS. Buildings with no result are drawn grey rather than left out, and the caption says how many they were: on a printed map a missing building looks exactly like open ground. The colour scale is clipped to the 2nd–98th percentile so a handful of extreme buildings cannot flatten every other one into a single shade. A quantity that is the same in every building — space heating during an August event, for instance — is dropped rather than drawn, and the caption names it: a panel of one flat colour looks like a measurement and is not one.
- **Both files come out of a microclimate event run too**, and both say so. An event covers a window of days, not a year, so the numbers are roughly a fiftieth of the annual ones: the picture is labelled `kWh/m² over the 8-day event` instead of `kWh/m²·yr`, `aggregate.json` carries an `energy_period` block naming the window, and every row in the layer carries `run_mode`, `event_days` and the temperature offset the building actually ran at (`delta_peak_k`) — so the map can be coloured on the cause as well as the consequence. Read the fields named per-year (`total_site_co2_t_yr`) as belonging to that window only. Only the buildings the slice reaches are in an event run at all, so the layer's extent is the slice's extent, not the city's.
- **Signed building package** — one building: its model, its preserved outputs, its ledger row, and a signed manifest. Use it to send a single result to someone who needs to check it.
- **Full signed ZIP** — the entire run. It first measures the real file count and uncompressed size and asks you to confirm, because a full-city package is large. The archive carries an Ed25519-signed manifest, so a recipient can verify nothing was altered in transit.

5 What the engine does with your files

Four files go in. Every building in scope gets its own model, its own annual simulation and its own row on disk. The rows are then added up. Nothing else happens.



There is no typological shortcut in this chain. A district result is the sum of that many individual annual simulations — not one representative building multiplied by a floor area.

Which source supplies which value, at every stage

Each stage is labelled with the file it consumes and the exact field it reads.

Building GIS · Tipo15 · Climate · Template

field names below are the real column names in the data

1 Footprint
The parcel polygon is simplified. If simplification moves its area too far, the building is excluded rather than distorted.
GIS · polygon geometry, `refparcela`

2 Storey count — capped by what the cadastre records
Height sets the storeys, but they are limited to what the recorded dwelling area can fill, plus one commercial ground floor.
GIS · `altura_max` → capped by Tipo15 · `442_sup_Residencial` summed over `31_pc`

3 Envelope, by typology and construction period
The building's TABULA class picks its walls, roof and floor — assembled layer by layer from real materials in the template.
GIS · `cluster` (21 classes) → Template · named constructions and materials

4 Windows, frames and blinds
Discrete openings per façade with balcony doors, aluminium frames and Valencian *persianas* that close on strong summer sun.
Template · glazing, frame and shading-control objects — placed on the geometry from stage 1

5 Neighbours — shading and party walls
Every real building within 50 m becomes shading mass; a shared wall becomes adiabatic instead of losing heat to the street.
GIS · neighbouring polygons and their heights, read from the untouched cadastre

6 Occupancy — the people actually registered there
The registered resident count replaces the standard 20 m²/person norm, changing both internal gains and hygienic ventilation.
GIS · `pob_total`, `num_vivend` → Template · CTE occupancy schedule kept unchanged

7 Domestic hot water
A hot-water loop sized on the Spanish CTE figure of 28 litres per person per day — so it follows the residents, not the floor area.
GIS · `pob_total` × 28 l/day → Template · boiler and loop objects

8 Heat pumps, sized on design days
One packaged terminal heat pump per zone, autosized against the extreme winter and summer days rather than a rule of thumb.
Climate · DDY winter + summer design days → Template · heat-pump performance curves

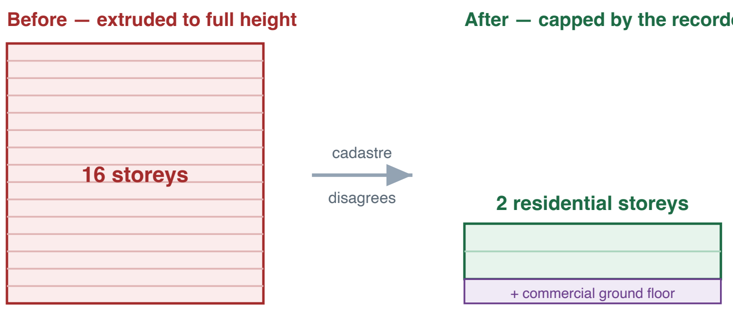
9 Annual simulation, then quality control
8,760 hours. The result is then checked back against the model's own geometry and the error file is scanned before the row is written.
Climate · EPW hourly year → checked against stages 1–4, then a row on disk

Stage 2 is the one that changed most recently, and it is the difference between conditioning 2.76 and 3.84 million m² across this district. It is drawn in §6.

6 Reading the numbers honestly

A parcel is not always one building

PARCEL 3748901YJ2734H · BENICALAP · BlocPluriP06 · altura_max = 15



17,273 m² footprint × 16
276,366 m² conditioned
5.6 % of the whole district's energy

342 dwellings · 38,158 m² recorded
51,819 m² conditioned
1.6 % of the district's energy

Across Benicalap

399 of 967 buildings were capped. They carry 55 % of the district's energy after the correction.

521 buildings came out identical to the bit — they were already short enough.

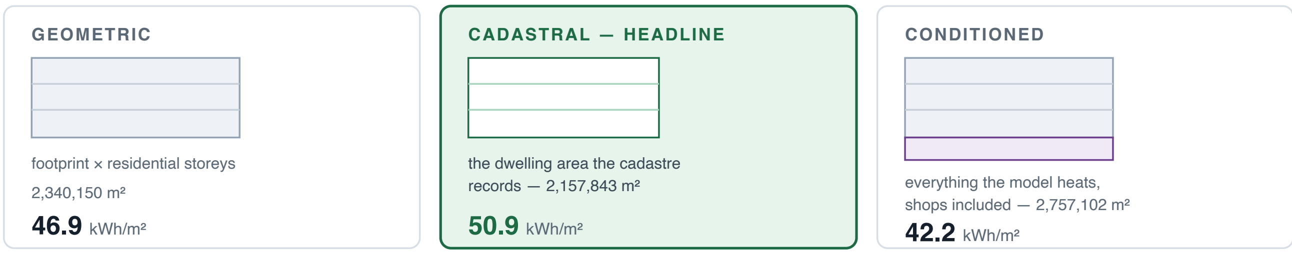
District total: 138.15 → 109.76 GWh

The footprint is the whole city block; the height is that of its tallest corner. Multiplying them invents floor area that does not exist, so the storeys are limited to what the cadastre can actually account for.

> **What the cap does not fix**
>
> A capped building gets the right *area* but keeps the full block footprint, so it is squat where the real building is a tower and its surface-to-volume ratio is wrong. Because heating and cooling are only 8.6 % of the total, the area correction dominates and the shape error is second-order — but it is real, and it stays on the list in §7.

Which floor area is the number divided by?

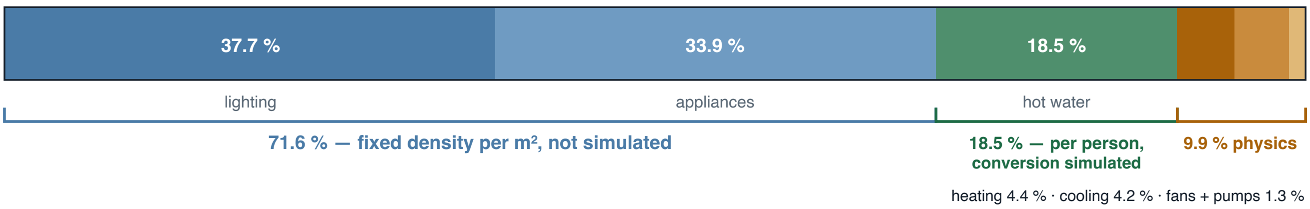
The same **109.76 GWh** divided three different ways. Every intensity in this product states which one it used.



▲ Same energy, three answers, a 20 % spread. A comparison that does not state the basis is not a comparison.

The headline uses the cadastral basis because that is the basis the published reference study uses — which is what makes the “Δ vs Rai” column on the Outputs screen a like-for-like comparison.

Where the energy actually comes from



Three kinds of number here, not two. Lighting and appliances are a fixed density per m²: a norm applied to floor area, never simulated. Hot water is normative only in its *draw* — a standard 28 litres per person per day applied to each building's real registered population — while the energy that heats those litres is genuinely simulated: a 97 % boiler, a 50 °C setpoint, mains temperature read from the climate file. Only the last band is envelope and HVAC physics: **work on insulation, shading or thermal bridges moves under a tenth of the headline** — which does not make it worthless, but does tell you what the headline is sensitive to.

How to check that we are not fooling ourselves

The tool rebuilds a building published by the reference author — from the surface areas he reported — and scores itself against his own results on thirteen quantities. Any failure makes the command exit non-zero.

QUANTITY	OURS	REFERENCE	DIFFERENCE	TOLERANCE
Residential floor area (m²)	954.800	954.800	0.00 %	0.5 %
Wall U-value (W/m²K)	1.409	1.409	0.00 %	1 %
Lighting (GJ)	65.308	65.320	−0.02 %	1 %
Hot water (GJ)	36.057	36.080	−0.06 %	1 %
HVAC sensible heating (GJ)	26.527	27.300	−2.83 %	5 %
HVAC sensible cooling (GJ)	−68.403	−65.030	+5.19 %	8 %
Total site energy (kWh/m²)	**53.680**	**54.500**	**−1.50 %**	3 %

Read this honestly

The reconstruction shares the reference author's template and weather file and is built from the surface areas he reported. It verifies the *engine and the parameter set*. It does not independently confirm that the typological abstraction, or the wider stock geometry, is right.

- **A profile fingerprint.** Every locked parameter and the checksum of every engine source file are hashed into one identifier (`80b3d4dafcdb7190…`) written into every result row. When the engine is later corrected the fingerprint is retired: the old result stays valid and traceable, but reproducing it byte-for-byte means checking out the source state it points to, not the newest version.
- **A drift guard.** If any engine source changes, a run refuses to start and names the parameter that moved — so no single run can span two versions of the physics.
- **Self-describing rows.** Each row carries the fingerprints of the inputs, the policy and the engine that produced it, so one row can be traced without the rest of the run.

7 Where this stands — my own notes

I would rather you heard this from me than found it yourself. Everything below is measured, not estimated, and the sizes are from the Benicalap run.

A **What I cannot do today**

- **I cannot tell you what a building actually consumes.** Nothing here has been compared with a meter. I report what the standards and the physics imply — not what happened. Affects every number in the product.
- **I cannot give you an uncertainty band.** The sampling study exists, but it was run on an earlier configuration of the engine, so the current figures carry no confidence interval. A range would have to be re-derived before I could quote one.
- **I cannot split heating and cooling between the dwellings and the shops.** EnergyPlus meters by end use, not by space type, so the residential subtotal I report is an *upper bound* rather than a clean split. Commercial ground floors are 10.0 % of site energy.
- **I cannot justify the carbon figure.** It rests on an assumed fuel mix for the real stock that nobody has confirmed for me yet. 34,723 tCO_2/yr — treat it as conditional.
- **I cannot recover the true shape of a capped block.** I get its area right and its massing only approximately; it stays squat where the real building is a tower. 399 buildings · 55 % of district energy.
- **I cannot resolve a deep floor plate properly.** Above a footprint threshold each storey is one well-mixed zone, with no core-versus-perimeter distinction. 9 buildings · 9.2 % of area · 8.0 % of energy — flagged per building, never silent.
- **I cannot model more than one weather year**, and I cannot model a warmed future climate. One measured year is used throughout. A single EPW file.
- **My totals cover the modelled subset only, and that subset is not a neutral sample.** Exclusions concentrate in large buildings, and larger footprints tend to have lower intensity — so the intensity I report is biased slightly upward relative to the full stock. 95.6 % of buildings · 98.1 % of footprint area.

B What could be better, but is not blocking

- **Lighting and appliances follow a fixed standard density.** That is 71.6 % of the site energy coming from a norm rather than a measurement. I keep it because the reference study uses the same assumption, which is what makes the comparison consistent — but consistency is not accuracy. No appliance survey exists for this stock.
- **The smallest model I can build is ground plus one storey**, so very short buildings cannot shrink any further even when the cadastre says they should. 74 buildings still condition more than twice their recorded dwelling area.
- **A building with no registered residents gets no hot water at all.** This is deliberate and matches the reference thesis; the alternative policy is implemented and can be selected instead. Policy is recorded in every row.
- **Some buildings have implausible population records** — far too many or too few people for their floor area. I flag them per building rather than correcting them, because the register is the source and I would rather you saw the anomaly than a quiet guess. 17 of 967 flagged in this district.
- **The envelope data covers only part of the stock.** The published reference models span roughly 31 % of Valencia's buildings; for the largest family — around 18,000 blocks, some 69 % of the stock — I am working from typological tables rather than a published model. I have asked Rai for the missing envelope definitions; that request is what closes this item.

C What I am working on right now

- **The full-city run.** The engine already screens 25,094 of the 26,452 parcels as buildable. I am deliberately not launching it until the envelope question above is settled — a five-day run on an assumption I expect to change is a waste of five days.
- **Extending the envelope data**, which is the item that gates the full-city run.
- **Transferring the method to a second city.** A full-stock run on Lecco completed and verified itself internally, but it has no external validation, so I label it a transferability scenario rather than a result — and I ran it from the command line, because the Workbench itself still has no city selector.

D What comes next, in order

1. **Settle the envelope question, then run the whole city.** Everything else is smaller than this.
2. **Put an uncertainty band on the numbers** by re-running the sampling study against the current configuration. Until then every figure is a point estimate with no stated spread.
3. **Separate HVAC by space type** so the residential subtotal becomes a real split rather than an upper bound. This needs an extra output table and a re-run.
4. **Calibrate against measured consumption.** This is the one that would change the standing of everything else — until measured data arrives, no number here can be presented as real consumption, only as what the standards and the physics imply.
5. **Make this work for any city, from the Files screen alone.** The climate side already does: upload any city's EPW and DDY, and the pair is validated, named from the file's own header, and its sizing days chosen by name. The rest is still Valencia-shaped — the companion file must carry the Spanish cadastral columns, the GIS must carry `refparcela` and `altura_max`, the reference column I compare against exists only for Valencia, and the third scope button still reads **All Valencia**. Two constants also travel unchanged into every climate bundle I build — ground temperature 18 °C and mains water 10 °C — because they were verified here and cannot be recovered from an EPW; for another climate they would have to be supplied, not inherited. The target is that loading four files is the whole job.

If you want to go past the three tabs

- **One building from the command line.** `python src/verified_model.py --refparcela <REFERENCE>` runs the identical pipeline for a single building. `--verify` re-runs the thirteen checks in §6; `--show-profile` prints every locked parameter.
- **Earlier research screens.** A geometry builder, a full OpenStudio editor, scenario controls, the representative-cluster pipelines and the sampling study are all still in the application, hidden from navigation but reachable by direct URL. They belong to the earlier method and are kept for reproducing historical results, not for new work.

TERM	MEANING
`refparcela`	The cadastral reference identifying a parcel. The primary key everywhere in this tool.
Cluster	A TABULA class combining building type with construction period P01–P07. Twenty-one exist in Valencia.
EUI	Energy Use Intensity — annual energy over floor area, in kWh/m². Meaningless unless the basis is stated (§6).
Tipo15	The cadastral companion file listing dwellings floor by floor: how many, what use, how much area.
`pob_total`	Registered residents per building, from the municipal register — actual residence, not design occupancy.
EPW / DDY	The annual hourly weather file, and the extreme design days used to size equipment.